Dr. Sontyana Adonijah Graham

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Education

Ph.D. Electronics and Information Convergence Engineering, Department of Electronics and Information Convergence Engineering, Institute for Wearable Convergence Engineering, Kyung Hee University, Republic of Korea. *2018-2023*
Thesis: A study on advanced materials and hybrid nanogenerators for sustainable energy harvesting and IoT applications.

M.Sc. Nanoscience and Nanotechnology, at the National Center for Nanoscience and Nanotechnology, University of Madras, Tamil Nadu, India. *2015-2017*

B.Sc. Nanoscience and Nanotechnology, Department of Nanotechnology, Acharya Nagarjuna University, Andhra Pradesh, India. *2012-2015*

Research Experience

- **Post-doctoral researcher**, Daegu Gyeongbuk Institute of Science and Technology (DGIST), & (KAIST) InnoCORE PRISM-AI Center, Korea Advanced Institute of Science and Technology Republic of Korea. *10.2025-ongoing*
  - *Develop applied IoT systems integrating AI for wearable and industrial applications.*
  - *Engineer triboelectric nanogenerators for self-powered IoT electronics.*
  - *Supervise MSc/PhD students, lead funding proposals, and collaborate across disciplines.*
- **Post-doctoral researcher**, Institute for Wearable Convergence Electronics (IWCE), Kyung Hee University, Republic of Korea. *09.2023 – 08.2024*
  - *Lead research on self-powered wearable electronics and IoT devices.*
  - *Integrate nanogenerators with artificial intelligence and biomedical systems.*
  - *Mentor multiple graduate students with tailored research directions.*
- **Ph.D. Researcher** Electronic Engineering at the Department of Electronic and Information Convergence Engineering, Kyung Hee University, Republic of Korea. *08.2018 - 08.2023*
  - *Synthesized and characterized advanced materials for energy harvesting, sensing, and electronics.*
  - *Integrated energy harvesters with power management, biomedical, and IoT systems.*
  - *Fabricated mechanical energy harvesters, analyzed performance, and mentored junior students; reviewed manuscripts for international journals.*
- **Researcher** at the "Targeted Imaging and Nanomedicine Laboratory, Department of Biomedical Science, Chonnam National University Medical School, Republic of Korea. *09.2017- 08.2018*

- *Investigated nanomaterials for biomedical imaging and therapy.*
- *Performed in vivo studies in mice using photothermal and electromagnetic treatments.*

- **Master's degree project** at the Microwave Laboratory, Department of Physics, Indian Institute of Technology (IIT) Madras, Tamil Nadu, India. *06.2016 - 04.2017*
  - *Developed bismuth ferrite thin films for multiferroic-based photovoltaic applications.*
  - *Employed thin-film deposition, thermal evaporation, and advanced characterization techniques.*
- **Internship** at the Department of Nanoscience and Nanotechnology, Bharathiar University, Tamil Nadu, India. *05.2016 - 06.2016*
  - *Hydrothermally derived different morphological tin oxide nanostructures coated on conductive textile for biosensing application.*
  - *Material and wearable fibers characterization techniques.*

Teaching and invited seminars

- **Inviter seminar:** 2026 International Forum on optoelectronic technology and optical engineering, Yangtze University, China. *22-24.05.2026*
  - Topic: Energy Materials and Triboelectric Nanogenerators for Energy Harvesting and Self-powered Electronics towards AI-Integrated IoT applications
- **Guest lecturer** for Master students: Acharya Nagarjuna University, India. *10.2025, 05.2026*
  - Topics: Advance Energy Materials, Energy harvesting and sensing devices, Nanotechnology for sustainable energy, Nanomaterials in Energy Conversion system, Electrochemical energy storage systems.
- **Invited Seminar**: Institute for Wearable Convergence Engineering, College of Electronics and Information. Kyung Hee University. Republic of Korea *18.03.2026*
  - Topic: Emerging DC-Generating TENG Mechanisms and their IoT applications
- **Invited Seminar**: Parousia International City School. Republic of Korea *15.01.2026*
  - Topic: AI and smart City (IoT, AI and data platforms, transportation, energy, healthcare, and safety)

Publication List

Selected First/co-first authored publications (14 published)

1. **Sontyana Adonijah Graham**, et. al. Metal-Organic Framework Embedded Electrospun Fibrous Membranes-Based Hybrid Nanogenerators with Hierarchical Modified Polyamide Films for Mechanical Energy Harvesting and IoT Applications. *Advanced Functional Materials* (*2025*) e07125, **I.F. 19** (doi.org/10.1002/adfm.202507125).
2. **Sontyana Adonijah Graham**, et. al. "Integrated design of highly porous cellulose-loaded polymer-based triboelectric films toward flexible, humidity-resistant, and sustainable

mechanical energy harvesters", ACS Energy Letters (*2020*) 5, 2140–2148. **I.F. 23.991.** (doi.org/10.1021/acsenergylett.0c00635)

3. **Sontyana Adonijah Graham**, et. al, "Multifunctional thermochromic dye-integrated hybrid nanogenerator for mechanical energy harvesting and real-time sensing", *Advanced Functional Materials* (*2024*) 2409608, **I.F. 19** (doi.org/10.1002/adfm.202409608)
4. **Sontyana Adonijah Graham**, et. al. "Engineering squandered cotton into eco-benign microarchitectured triboelectric films for sustainable and highly efficient mechanical energy harvesting", Nano Energy (*2019*) 61, 505-516. **I.F. 19.069** (doi.org/10.1016/j.nanoen.2019.04.081)
5. **Sontyana Adonijah Graham**, et. al. "Triboelectric charge modulation to understand the electrification process in nanogenerators combined with an efficient power management system for IoT applications", Nano Energy (*2023*) 111, 108413, **I.F. 19.069** (doi.org/10.1016/j.nanoen.2023.108413)
6. **Sontyana Adonijah Graham**, et. al. "Biocompatible electrospun fibers-based triboelectric nanogenerator for energy harvesting and healthcare monitoring", *Nano Energy* (*2022*) 100, 107455. **I.F. 19.069** (doi.org/10.1016/j.nanoen.2022.107455)
7. **Sontyana Adonijah Graham**, et. al. "Harsh environment–tolerant and robust triboelectric nanogenerators for mechanical-energy harvesting, sensing, and energy storage in a smart home", *Nano Energy* (*2021*) 80, 105547. **I.F. 19.069** (doi.org/10.1016/j.nanoen.2020.105547)
8. Bhaskar Dudem, **Sontyana Adonijah Graham**, et. al. "Natural silk-composite enabled versatile robust triboelectric nanogenerators for smart applications", *Nano Energy* (*2021*) 83, 105819. **I.F. 19.069,** (Equal first authorship). (doi.org/10.1016/j.nanoen.2021.105819)

Selected Co-authored/Collaboration publications (47 published)

1. Seneke Chamith Chandrarathna, **Sontyana Adonijah Graham**, et. al. "An Efficient Power Management System Using Dynamically Configured Multiple Triboelectric Nanogenerators and Dual-Parameter Maximum Power Point Tracking", *Advanced Energy Materials* (*2022*) 12.2, 2103249. **I.F. 29.698** (doi.org/10.1002/aenm.202103249)
2. Harishkumarreddy Patnam, **Sontyana Adonijah Graham**, et. al. "Highly-flexible and harsh temperature-tolerant single-electrode mode triboelectric nanogenerators via biocompatible ionic liquid electrolytes for wearable electronic applications", *Advanced Composite Hybrid Materials* (*2024*) 7(2), 5, **IF: 20.1** (doi.org/10.1007/s42114-024-00845-2)
3. Mandar Vasant Paranjape, Jun Kyu Lee, Punnarao Manchi, **Sontyana Adonijah Graham**, et. al. "Phosphor-Loaded Triboelectric Film-Based Multipurpose Triboelectric Nanogenerators for Highly-Efficient Energy Harvesting, Sensing, and Self-Illumination Applications", *Advanced Functional Materials* (*2024*) 2405838. **IF: 19** (doi.org/10.1002/adfm.202405838)
4. Paranjape Mandar Vasant, **Sontyana Adonijah Graham**, et. al. "Microarchitectured Strontium Doped Silver Niobate Embedded Ecoflex Composite Films for Highly Efficient Box-type Mechanical Energy Harvesters", *Nano Energy* (*2023*) 109005. **I.F. 19.069** (doi.org/10.1016/j.nanoen.2023.109005)

5. Chandrarathna Seneke Chamith, **Sontyana Adonijah Graham**, et. al. "Analysis and Experiment of Self-Powered, Pulse-Based Energy Harvester Using 400 V FEP-Based Segmented Triboelectric Nanogenerators and 98.2% Tracking Efficient Power Management IC for Multi-Functional IoT Applications", *Advanced Functional Materials* (*2023*), 33.17, 2213900. **I.F. 19.924** (doi.org/10.1002/adfm.202213900)

Technical Skills

- **Processing techniques** → Patterning: Soft imprint lithography, photolithography, thermal imprint lithography. → Vacuum deposition techniques: RF/DC magnetron sputtering, plasma-enhanced chemical vapour deposition, electron-beam, and thermal evaporation techniques. – Chemical Synthesis: Chemical bath deposition, co-precipitation, anodization, electro-deposition, electro-polishing. – Etching: Inductively coupled plasma etching and wet chemical etching.
- **Characterization Techniques** → Material characterization: Scanning electron microscope (SEM), atomic & kelvin probe force microscopy (AFM & KPFM), Raman spectrometer, UV-vis-NIR spectrophotometer, 3D laser scanning, X-ray diffraction (XRD), Fourier transformer infrared spectroscopy (FTIR), dielectric measurements, ferroelectric, X-ray photoelectron spectroscopy (XPS).
- Confocal microscope. → Electrical measurements: Oscilloscopes (Tektronix), Current preamplifier, and Keithley (6514) electrometers were used to examine the output performance of nanogenerators and sensors.
- **Software Skills** → Simulation software: Finite Element Simulation tool (COMSOL Multiphysics). 3D modelling and Data analysis software: Autodesk 3ds Max, Fusion; Origin; etc.

Selected Conference Presentations (12 presented)

1. **Sontyana Adonijah Graham**, J.H. Lee. "Virbo-active MOF-based hybrid nanogenerator for ambient mechanical energy harvesting and self-powered vibration sensing" NGPT2026 8th International Conference on Nanogenerator and Piezotronics" Sungkyunkwan University, Seoul, Korea *12-06-2026.* (**Poster presentation**)
2. **Sontyana Adonijah Graham**, J.S. Yu. "Multiferroic-based triboelectric nanogenerator for energy harvesting and sensing applications" Nano Korea 2023 Symposium, Kintex, Korea, *5-07-2023*. (**Oral presentation**)
3. **Sontyana Adonijah Graham**, J.S. Yu. "Synthesis of bismuth composite-based hybrid metamaterials for discrete energy harvesting" *European Material Research Society in Warsaw*, Poland, *22-09-2022*. (**Oral presentation**).
4. **Sontyana Adonijah Graham**, J.S. Yu. "Electrospun biocompatible fibers as a tribosensor for human body motion" *The 20th International Symposium on the Physics of Semiconductors and Applications*, Jeju, Korea*, 21-07-2022*. (**Oral presentation**).
5. **Sontyana Adonijah Graham**, J.S. Yu. "Enhanced surface area triboelectric films for wind energy harvesting rotatory triboelectric nanogenerator" *The 22nd International Union of Materials Research Societies*, Jeju, Korea, *8-10-2021*. (**Oral presentation**).

6. **Sontyana Adonijah Graham**, J.S. Yu. "Dynamic transformation of ubiquitous cotton fibers for eco-friendly energy conversion systems" Workshop-Center for the Smart Energy Platform, Kyung Hee University, Korea, *10-06-2019*. (**Oral presentation**).

Selected International and National Awards and Achievements (7 awards)

1. **Seal of excellence,** MSCA Postdoctoral Fellowship 2025, was recognized as a high-quality project proposal in a highly competitive evaluation process. European Commission Horizon Europe, Europe (*16-02-2026*)
2. **Graduate student award**, European - Material Research Society (E-MRS) 2022 Fall Meeting, Warsaw University of Technology, Poland (*22-09-2022*)
3. **Excellent patent award**, BK21 Convergence Future Communication Innovation Talent Cultivation. Kyung Hee University, Republic of Korea (*19-11-2021*)
4. **Best presentation award**, International Virtual Conference on Nanomaterials, India. (*Sep. 2020*)
5. **Best poster presentation award**, Korean Vacuum Society, Republic of Korea (*12-08-2019*)

Selected Patents (4 published and 4 under review)

1. Cotton-Based Biodegradable Nano Power Generation Device, Org. Korean Intellectual Property Office. Ref. No. 10-2020-0144708, Approved date: *19-06-2019*, *Inventor names: Jae Su Yu,* ***Sontyana Adonijah Graham****, and Dudem Bhaskar*
2. Smart Home Mechanical Energy Harvesting, Sensing and Energy Storage by Cotton-Based Triboelectric Nanogenerator, Org. Korean Intellectual Property Office. Ref. No. 10-2022-0119871, Approved date: *22-02-2021*, *Inventor names: Jae Su Yu, and* ***Sontyana Adonijah Graham***.
3. Micropatterning to Release High Surface Area Triboelectric Films for Multimode Operation Triboelectric Nanogenerator, Org. Korean Intellectual Property Office. Ref. No. 10-2023-0085496, Approved date: *07-12-2021*. *Inventor names: Jae Su Yu, and* ***Sontyana Adonijah Graham***.

Media Coverage

- New Self-powered Energy Supply Device Harvests Electricity during Everyday Activities. Kyung Hee Univesity Focus. *07-09-2020*. Web link: Kyung Hee University (khu.ac.kr).
- Harvesting Electric Energy from Physical Exercise in Everyday Life. Kyung Hee Univesity Focus. *07-02-2022*. Web link: Kyung Hee University (khu.ac.kr).
- Development of eco-friendly, high-efficiency triboelectric nano power generation device. Kyung Hee Univesity Focus. *17-07-2023* Web link: Kyung Hee University (khu.ac.kr).

Reference

- Will be provided upon request.